12th HACKER JUGAAD NOTES

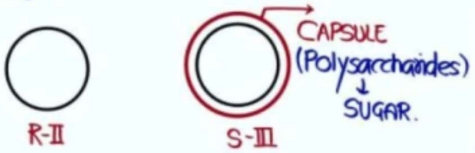
Molecular basis of Inheritance

DNA IS A GENETIC MATERIAL

F. GRIFFITH EXPERIMENT, 1928

★ STREPTOCOCCUS PNEUMONIAE

★ R-II strain, S-III strain → ② FORM



	R-II	S-III
Colony/cell	ROUGH (NOT SHINING)	SMOOTH APPEARANCE (SHINING)
CAPSULE	ABSENT	PRESENT
	NON-PATHOGENIC/ NON VIRULENT	PATHOGENIC/ VIRULENT
PNEUMONIA	NO	YES

MICE + R-II (GOOD) → NO PNEUMONIA → NO DEATH.

MICE + S-III (BAD) → PNEUMONIA → DEATH.

MICE + S-III (HEAT KILLED) → NO PNEUMONIA → NO DEATH.

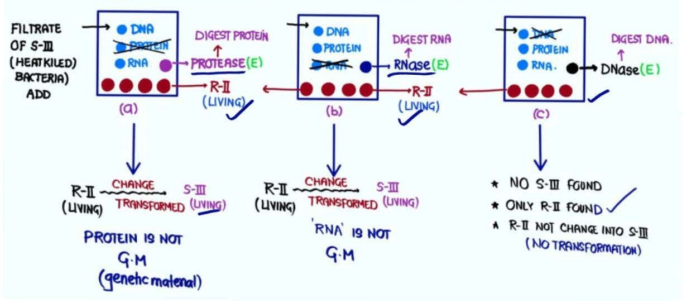
MICE + R-II (GOOD) + S-III (HEAT KILLED) → DEATH.

HE WAS EXPECTING 'NO DEATH' BUT MICE DIED

GRIFFITH SAID: SOMETHING (TRANSFORMING PRINCIPLE/ GENETIC MATERIAL) TRANSFER FROM HEAT KILLED S-III BACTERIA TO LIVING R-II BACTERIA SO R-II DEVELOP CAPSULE AND CHANGE INTO LIVING S-III BACTERIA WHICH KILLED MICE.

★ WHAT IS 'BIOCHEMICAL COMPOSITION OF 'SOMETHING' NOT EXPLAINED BY GRIFFITH.

BIOCHEMICAL CHARACTERISATION OF 'TRANSFORMING PRINCIPLE' — Avery, Macleod, McCarty



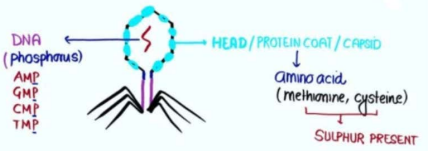
HERSHEY AND CHASE EXP. 1952

★ ONEQUIVOCAL PROOF: DNA IS GENETIC MATERIAL

★ VIRUS: T_2 → infection BACTERIA (BACTERIOPHAGE), ds DNA

★ E. COLI ✓

★ CONCLUSION: 'P': DNA, 'S': PROTEIN COAT



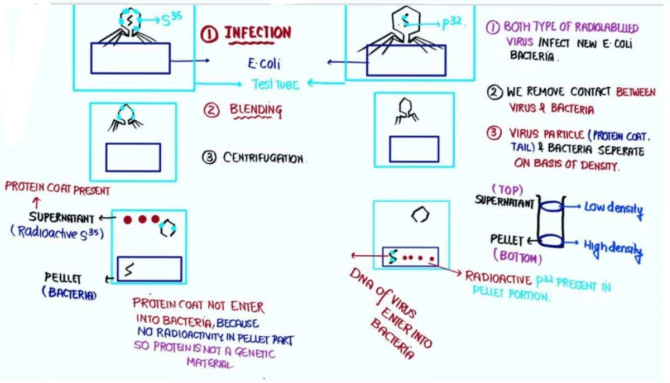
★ SOME T-2 VIRUS GROW IN A MEDIUM CONTAINING RADIOACTIVE S^{35} → ENTER INTO PROTEIN COAT

★ SOME T-2 VIRUS GROW IN A MEDIUM CONTAINING RADIOACTIVE P^{32}



EXPERIMENT

① INFECTION
② BLENDING
③ CENTRIFUGATION

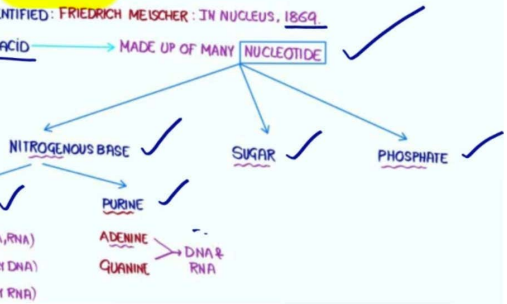


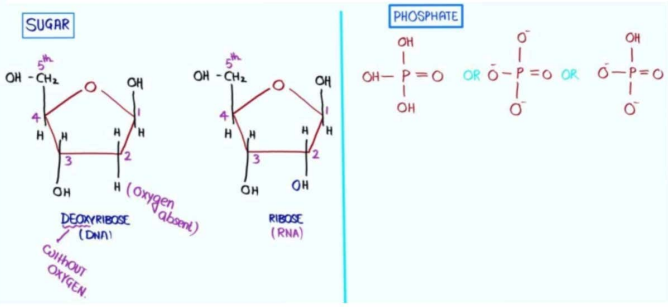
DNA

(PHOSPHATE)

★ ACIDIC SUBSTANCE: IDENTIFIED: FRIEDRICH MEISCHER: IN NUCLEUS, 1869

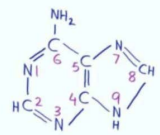
★ DNA → NUCLEIC ACID → MADE UP OF MANY NUCLEOTIDE ✓



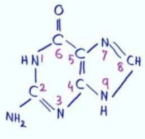


N_2 BASE

PURINE



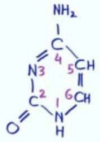
ADENINE



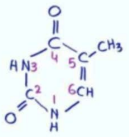
GUANINE

★ HETROCYCLIC COMPOUND
★ DOUBLE RING
★ NITROGEN POSITION: 1, 3, 7, 9.

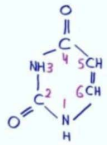
PYRIMIDINE



Cytosine



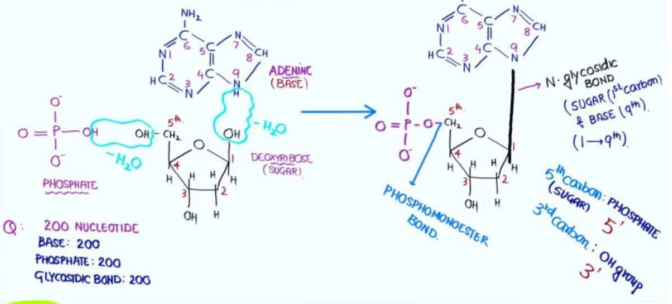
Thymine (5-METHYL URACIL)



URACIL

★ HETROCYCLIC COMPOUND
★ SINGLE RING
★ 'N': 1st, 3rd.

NUCLEOTIDE AND BONDING (ADENINE/GUANINE) (PURINE).



Q: 200 NUCLEOTIDE
BASE: 200
PHOSPHATE: 200
GLYCOSIDIC BOND: 200

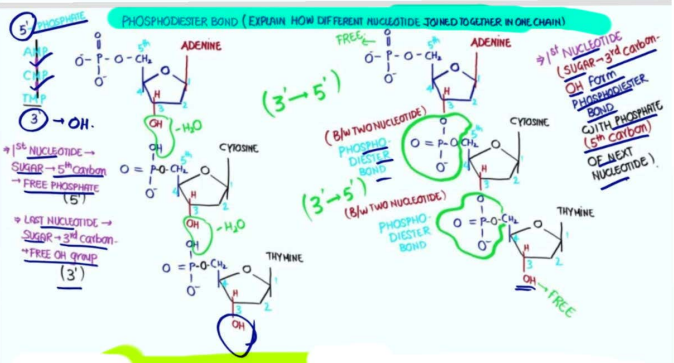
DNA

BASE	SUGAR	NUCLEOSIDE	+ PHOSPHATE	NUCLEOTIDE
ADENINE +	DEOXYRIBOSE →	DEOXYADENOSINE	PHOSPHATE	DEOXYADENOSINE MONO PHOSPHATE (AMP) / DEOXYADENYLIC ACID
GUANINE +	DEOXYRIBOSE →	DEOXYGUANOSINE	PHOSPHATE	DEOXYGUANOSINE MONO PHOSPHATE (GMP) / DEOXYGUANYLIC ACID
THYMINE +	DEOXYRIBOSE →	DEOXYTHYMIDINE	PHOSPHATE	DEOXYTHYMIDINE MONO PHOSPHATE (TMP) / DEOXYTHYMIDYLIC ACID
CYTOSINE +	DEOXYRIBOSE →	DEOXYCYTIDINE	PHOSPHATE	DEOXYCYTIDINE MONO PHOSPHATE (CMP) / DEOXYCYTYDYLIC ACID

RNA

ADENINE +	RIBOSE →	ADENOSINE	PHOSPHATE →	ADENOSINE PHOSPHATE
GUANINE +	RIBOSE →	GUANOSINE	PHOSPHATE →	GUANOSINE PHOSPHATE
URACIL +	RIBOSE →	URIDINE	PHOSPHATE →	URIDINE PHOSPHATE
CYTOSINE +	RIBOSE →	CYTIDINE	PHOSPHATE	CYTIDINE PHOSPHATE

PHOSPHODIESTER BOND (EXPLAIN HOW DIFFERENT NUCLEOTIDE JOINED TOGETHER IN ONE CHAIN)

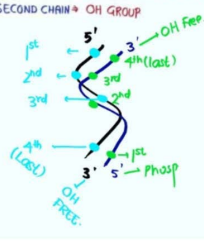


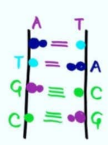
STRUCTURE OF DNA

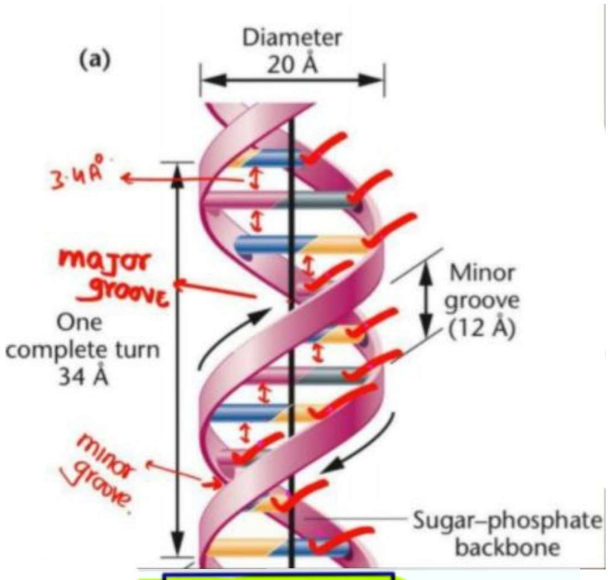
★ FRANKLIN & WILKIN → X RAY FALL ON DNA FIBRE → X RAY REFLECTED → PATTERN STUDY, X-RAY DIFFRACTION PATTERN OF DNA (X-RAY CRYSTALLOGRAPHY)
DNA IS A HELICAL STRUCTURE
BUT THEY DIDN'T PROVIDE PROPER DNA STRUCTURE

★ TWO TYPE OF STUDY - HELP → WATSON & CRICK → TO GIVE DETAIL STRUCTURE OF DNA, 1953
a) FRANKLIN & WILKIN ✓
b) CHARGAFF RULE ✓

① DNA made up of TWO POLYNUCLEOTIDE CHAIN
② BOTH CHAIN ARE ANTIPARALLEL.
(ONE CHAIN → PHOSPHATE AT ONE END
SECOND CHAIN → OH GROUP







WATSON & CRICK STRUCTURE

★ B-TYPE OF DNA, RIGHT HANDED DNA / CLOCKWISE
★ LENGTH OF DNA / HELIX LENGTH / PITCH OF DNA =
no. of Base pair × distⁿ B/w Two Bp (vertical RISE PER BP)
➡ 10 Base pair × 3.4 Å / 0.34 nm
⇒ **34 Å / 3.4 nm** : 1 TURN OF DNA (HELIX LENGTH)

★ LENGTH OF HUMAN DNA:
no. of Base pair × distⁿ B/w Two BP
⇒ $6.6 \times 10^9 \times 0.34 \times 10^{-9}$ m
⇒ **2.2 m**

★ DIAMETER: 20 Å
★ DISTANCE B/W TWO CHAIN IS UNIFORM BECAUSE PURINE FORMS BOND WITH PYRIMIDINE AND VICE-VERSA.

⇒ COMPLEMENTARY BASE PAIRING: UNIQUE PROPERTY OF DNA OR HALLMARK ✓
⇒ STABILITY OF DNA: HYDROGEN BONDING B/W BASE PAIR, BASE STACKING (ONE BASE PAIR ABOVE OTHER)
⇒ BACK BONE: SUGAR ↔ PHOSPHATE (PHOSPHODIESTER BOND)
⇒ BASE PROJECTED INSIDE

CHARGAFF RULE (VALID FOR DS DNA)

a) PURINE = PYRIMIDINE
$\frac{A+G}{C+T} = 1$ ✓
$\frac{A+G}{C+T} = 1$
b) $\frac{A+T}{G+C}$ Ratio specific for species
c) Adenine equal to THYMINE
d) Guanine equal to cytosine
e) A = T (2 Hydrogen BOND)
f) G ≡ C (3 Hydrogen BOND)
g) SUGAR = PHOSPHATE

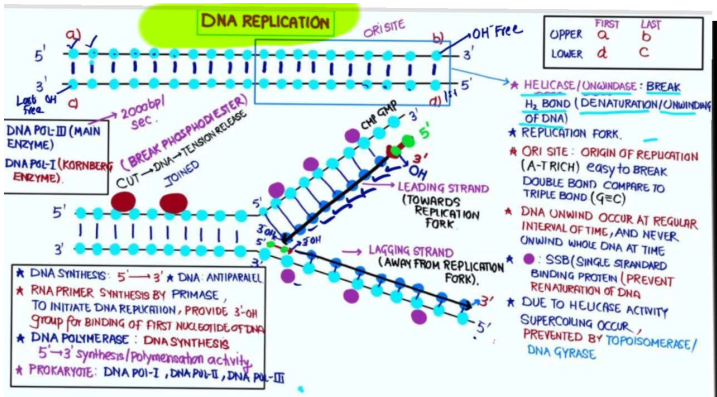
NOTE

Ø x 174 ⇒ 5386 base/NUCLEOTIDE, SS CIRCULAR DNA. (Length equal to number of Base)
λ PHAGE ⇒ 48502 basepair, dS Linear DNA
E. Coli ⇒ 4.6×10^6 bp, Haploid.
HUMAN (DIPLOID) ⇒ 6.6×10^9 bp ⇒ 46 chromosome
HUMAN (genome) Haploid set of ⇒ 23 chromosome
3.3×10^9 bp

CENTRAL DOGMA



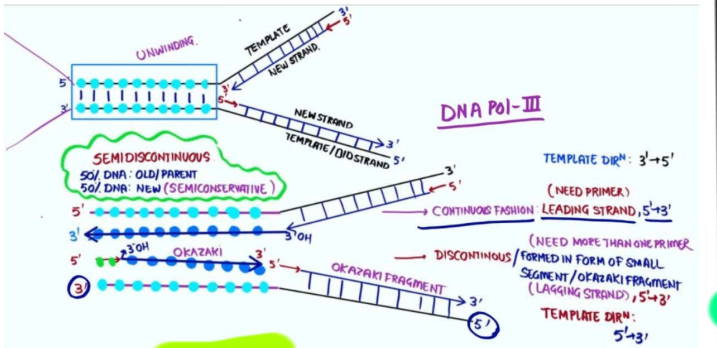
DNA REPLICATION



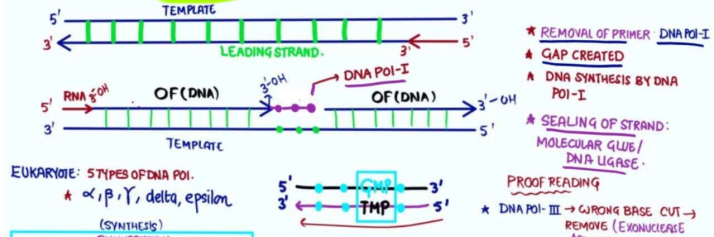
	FIRST	LAST
UPPER	a	b
LOWER	d	c

★ HELICASE / UNWINDASE : BREAK H_2 BOND (DENATURATION / UNWINDING OF DNA)
★ REPLICATION FORK.
★ ORI SITE : ORIGIN OF REPLICATION (A-T RICH) easy to break DOUBLE BOND COMPARE TO TRIPLE BOND (G≡C)
★ DNA UNWIND OCCUR AT REGULAR INTERVAL OF TIME, AND NEVER UNWIND WHOLE DNA AT TIME
★ ● : SSB (SINGLE STRANDARD BINDING PROTEIN (PREVENT RENATURATION OF DNA
★ DUE TO HELICASE ACTIVITY SUPERCOILING OCCUR, PREVENTED BY TOPOISOMERASE / DNA GYRASE

★ DNA SYNTHESIS : 5'→3' ★ DNA : ANTIPARALEL
★ RNA PRIMER SYNTHESIS BY PRIMASE, TO INITIATE DNA REPLICATION, PROVIDE 3'-OH Group for BINDING OF FIRST NUCLEOTIDE OF DNA
★ DNA POLYMERASE : DNA SYNTHESIS 5'→3' synthesis / polymerization activity.
★ PROKARYOTE : DNA POL-I, DNA POL-II, DNA POL-III



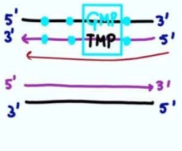
CLIMAX



★ REMOVAL OF PRIMER : DNA POL-I
★ GAP CREATED
★ DNA SYNTHESIS BY DNA POL-I
★ SEALING OF STRAND : MOLECULAR GLUE / DNA LIGASE.

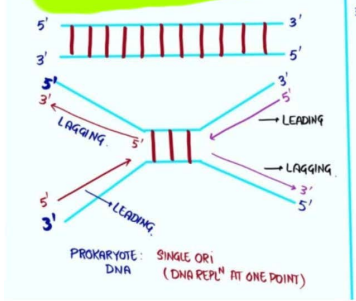
EUKARYOTE : 5 TYPES OF DNA POL.
★ α, β, γ, delta, epsilon

(SYNTHESIS)	POLYMERISATION ACTIVITY	EXONUCLEASE ACTIVITY
DNA POL-I	5'→3'	5'→3' & 3'→5'
DNA POL-II	5'→3'	3'→5'
DNA POL-III (main)	5'→3'	3'→5'

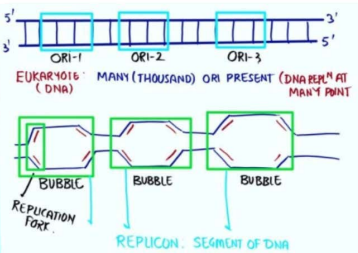


PROOF READING
★ DNA POL-III → WRONG BASE CUT → REMOVE (EXONUCLEASE ACTIVITY)
5'→3' exon activity
3'→5' exonuclease activity

NOTE : DNA REPLICATION : BIDIRECTIONAL



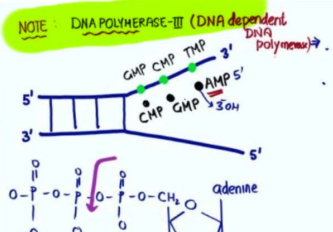
PROKARYOTE : SINGLE ORI (DNA REPL^N AT ONE POINT)



EUKARYOTE : MANY (THOUSAND) ORI PRESENT (DNA REPL^N AT MANY POINT)

REPLICON : SEGMENT OF DNA WHICH PARTICIPATE IN DNA REPLICATION.

NOTE : DNA POLYMERASE-III (DNA dependent DNA polymerase)



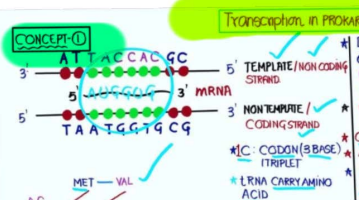
NUCLEOTIDE : DUAL FUNCTION
→ provide 3'-OH group for BINDING OF NEXT NUCLEOTIDE
→ 2 Phosphate (energy) Release. FORM PHOSPHODIESTER BOND B/W TWO NUCLEOTIDE

PRIMASE : RNA SYNTHESIS (PRIMER) DNA DEPENDENT RNA POLYMERASE

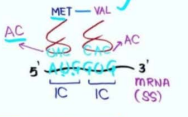
ATP → AMP + PP_i (Pyrophosphate)
GTP → GMP + PP_i
CTP → CMP + PP_i

Transcription in PROKARYOTES

CONCEPT-1



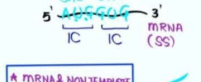
★ IC : CODON (3 BASE) (TRIPLET)
★ tRNA CARRY AMINO ACID
★ AC : ANTICODON
★ — : PEPTIDE BOND
★ MET → VALINE (amino acid chain) → FORM PROTEIN.



★ mRNA & NON TEMPLATE SEQUENCE ARE SAME (EXCEPT URACIL)
★ mRNA & TEMPLATE : COMPLEMENTARY.

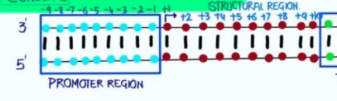
DNA → mRNA SYNTHESIS (5'→3') ★ add^n of RIBONUCLEOTIDE OR RIBONUCLEOSIDE TRIPHOSPHATE (ANP, CMP, UMP, GMP) ★ PP_i Released helps in PHOSPHODIESTER BOND. ★ RNA POLYMERASE (RNA SYNTHESIS) / DNA DEPENDENT RNA POLYMERASE → BREAKS H_2 BOND
ONLY ONE STRAND OF DNA PARTICIPATE. BUT IN REPLICATION : BOTH STRAND PARTICIPATE ★ NO PRIMER REQUIRE AS IN DNA REPL^N.
ONLY A PART OF TEMPLATE PARTICIPATE ★ mRNA CODE FOR PROTEIN
AUG : METHIONINE ; GUG : VALINE
MESSENGER RNA CARRY MESSAGE FROM DNA IN FORM OF TRIPLET / CODON AND THIS CODON READ BY PARTICULAR tRNA WHICH CARRY AMINO ACID. SO HELPS IN PROTEIN FORMATION.

CONCEPT-2 : GENE BOTH STRAND DO NOT PARTICIPATE

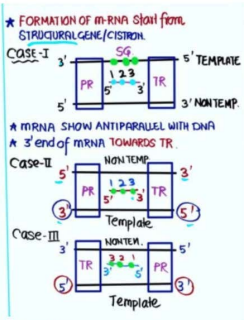


BOTH mRNA SEQUENCE : COMPLEMENTARY SO THEY FORM H_2 BOND → mRNA BECOMES DOUBLE STRAND → SO NO tRNA BINDING → NO PROTEIN SYNTHESIS.
→ CODON OF BOTH RNA DIFFERENT SO THEY BRING DIFF. AMINO ACID SO TWO DIFFERENT TYPE OF PROTEIN, SO IT WILL COMPLICATE THE MACHINERY

CONCEPT-3 : TRANSCRIPTIONAL UNIT

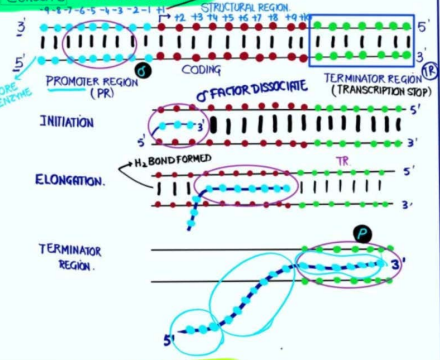


★ FORMATION OF m-RNA start from STRUCTURAL GENE / CISTRON.



★ mRNA SHOW ANTIPARALLEL WITH DNA
★ 3' end of mRNA TOWARDS TR.

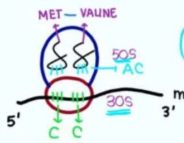
CONCEPT-3 : TRANSCRIPTIONAL UNIT



RNA POLYMERASE (CORE ENZYME)
★ + σ → RNA POLYMERASE (HOLOENZYME) (SIGMA FACTOR)
★ RNA POL (CORE ENZYME) BINDS TO PR WITH HELP OF σ FACTOR / PROTEIN
★ RNA POL BINDS TO PARTICULAR SEQ. IN PR CALLED CONSENSUS SEQUENCE (TATAAT → PRIBNOW BOX) & TTGACA
UPSTREAM | +1 | DOWNSTREAM
PR, 5'end | TR, 3'end
★ IF σ FACTOR ABSENT : TRANSCRIPTION OCCUR (NON-SPECIFIC)
★ RHO(P) PROTEIN BINDS TO CORE ENZYME AND RECOGNISE TR

PROKARYOTES

★ ONLY ONE TYPE OF RNA POLYMERASE
mRNA → PROVIDE CODON
tRNA → CARRY AMINO ACID
rRNA → IN RIBOSOME
→ ALL TYPE OF RNA NEED FOR PROTEIN SYNTHESIS (TRANSLATION)



CATALYTIC ROLE
STRUCTURAL ROLE
★ IT HELPS IN BINDING OF mRNA TO RIBOSOME (HOW ?)

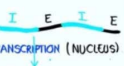
★ C : CODON.
A AC : ANTICODON
★ — : peptide BOND FORMED BY r-RNA (CATALYTIC ROLE) ↓ ACT AS enzyme (RIBOZYME)

S-G / CISTRON
CISTRON
★ SEGMENT OF DNA which CODE FOR PROTEIN / CHAIN OF AMINOACID / POLYPEPIDE
DNA → mRNA → PROTEIN.

TRANSCRIPTION IN EUKARYOTE

★ RNA POLYMERASE BINDS TO PARTICULAR SEQUENCE IN PROMOTER REGION → TATA BOX (TATATAT)

SPLIT GENE
E + I



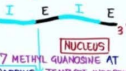
DNA / STRUCTURAL GENE / CISTRON

TRANSCRIPTION (NUCLEUS)



(PRIMARY TRANSCRIPT)
IMMATURE m-RNA
hn RNA (Heterogenous nuclear RNA)

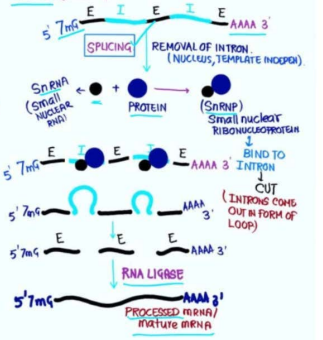
NUCLEUS
Addition of 7 METHYL GUANOSINE AT 5' END : CAPPING
TEMPLATE INDEPENDENT
GUANYL TRANSFERASE

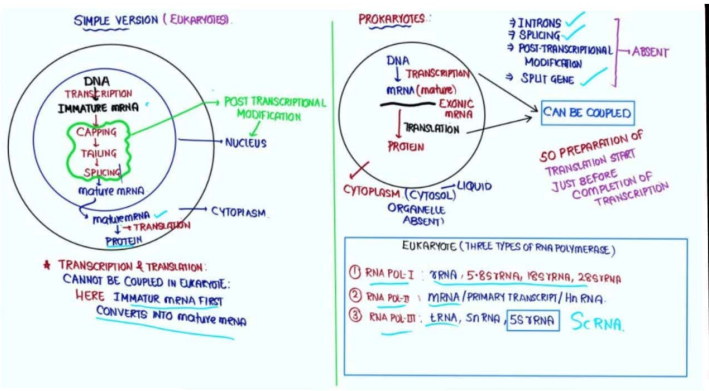


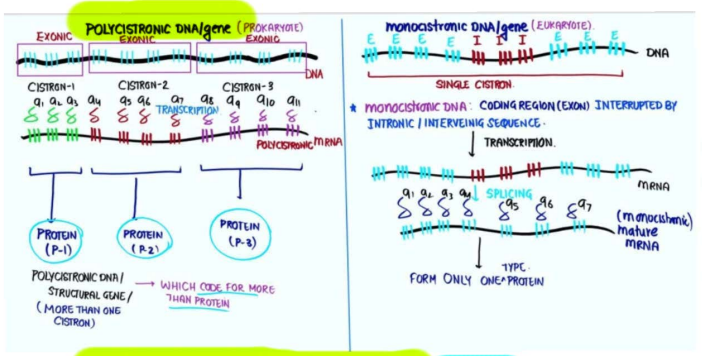
PREVENT DEGRADATION OF m-RNA FROM 5' end

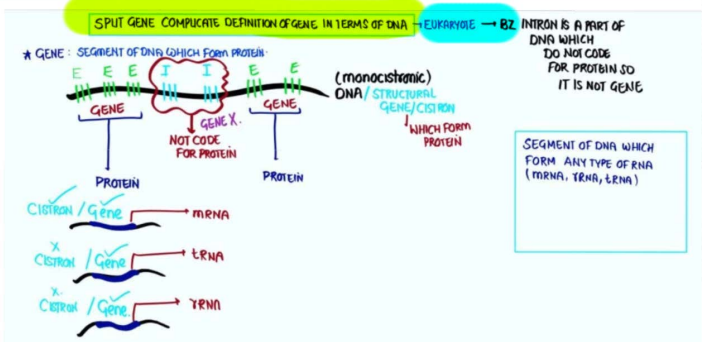
Addition of adenylic acid (200 to 300) at 3' end : TAILING / POLYADENYLATION
TEMPLATE INDEPENDENT (NUCLEUS)
POLY A POLYMERASE

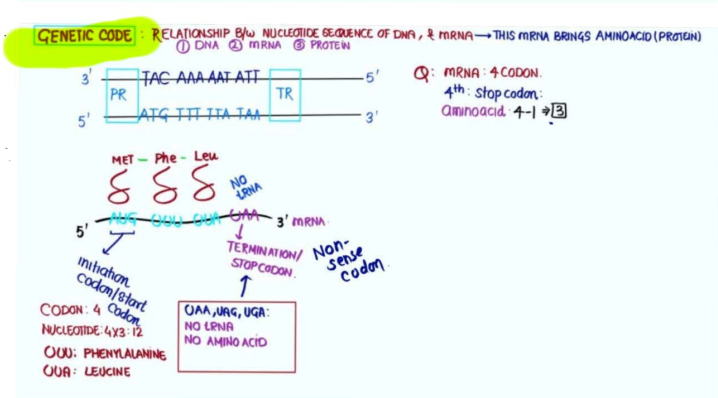
>95% (INTERVENING SEQUENCE)
I : INTRON
E : EXON
INTRON : NON CODING PART (NOT FORM PROTEIN)
EXON : CODING PART (FORM PROTEIN)
2%

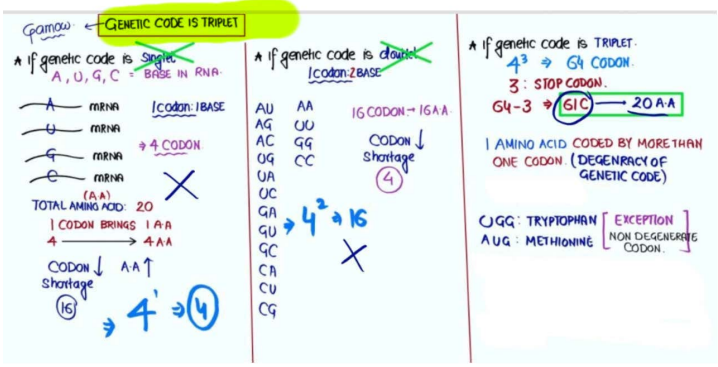


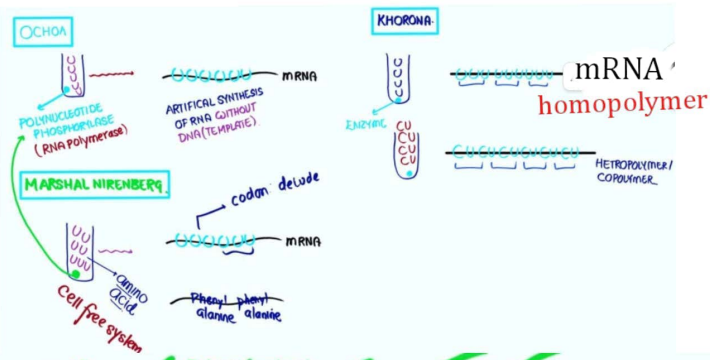

SIMPLE VERSION (EUKARYOTES)
DNA
TRANSCRIPTION
IMMATURE mRNA
CAPPING
TAILING
SPLICING
mature mRNA
POST TRANSCRIPTIONAL MODIFICATION
NUCLEUS
CYTOPLASM
mature mRNA
TRANSLATION
PROTEIN
* TRANSCRIPTION & TRANSLATION: CANNOT BE COUPLED IN EUKARYOTE; HERE IMMATURE mRNA FIRST CONVERTS INTO mature mRNA
PROKARYOTES:
DNA
TRANSCRIPTION
mRNA (mature)
EXONIC mRNA
TRANSLATION
PROTEIN
CYTOPLASM (CYTOSOL)
LIQUID
ORGANELLE ABSENT
INTRONS
SPLICING
POST-TRANSCRIPTIONAL MODIFICATION
SPLIT GENE
ABSENT
CAN BE COUPLED
SO PREPARATION OF TRANSLATION START JUST BEFORE COMPLETION OF TRANSCRIPTION
EUKARYOTE (THREE TYPES OF RNA POLYMERASE)
① RNA POL I : rRNA, 5·8S rRNA, 18S rRNA, 28S rRNA
② RNA POL II : mRNA / PRIMARY TRANSCRIPT / Hn RNA.
③ RNA POL III : tRNA, SnRNA, 5S rRNA, Sc RNA.

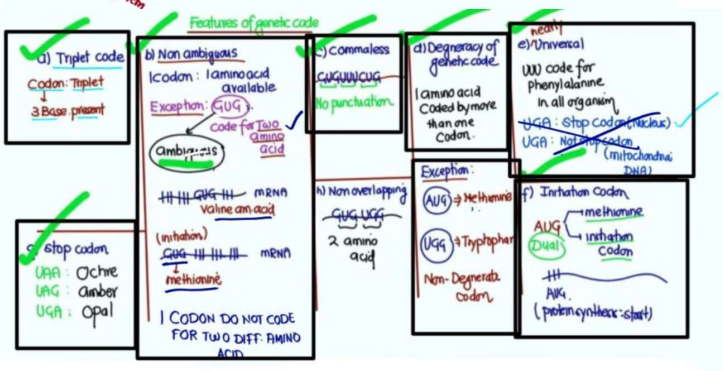

POLYCISTRONIC DNA/gene (PROKARYOTE)
EXONIC
DNA
CISTRON-1
CISTRON-2
CISTRON-3
TRANSCRIPTION
POLYCISTRONIC mRNA
PROTEIN (P-1)
PROTEIN (P-2)
PROTEIN (P-3)
POLYCISTRONIC DNA / STRUCTURAL GENE / (MORE THAN ONE CISTRON)
WHICH CODE FOR MORE THAN PROTEIN
monocistronic DNA/gene (EUKARYOTE)
SINGLE CISTRON
* monocistronic DNA: CODING REGION (EXON) INTERRUPTED BY INTRONIC / INTERVENING SEQUENCE
TRANSCRIPTION.
mRNA
SPLICING
(monocistronic) mature mRNA
TYPE.
FORM ONLY ONE PROTEIN

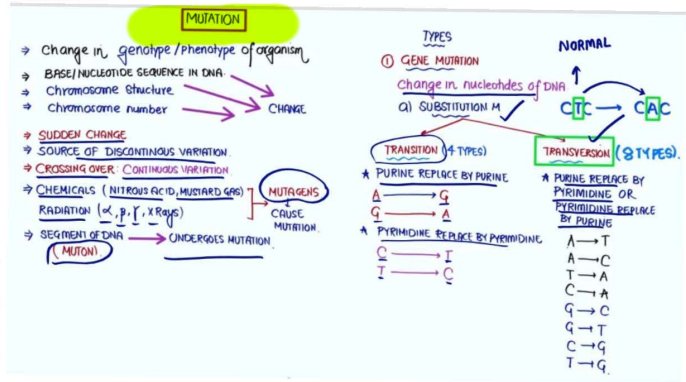

SPLIT GENE COMPLICATE DEFINITION OF GENE IN TERMS OF DNA
EUKARYOTE → 82. INTRON IS A PART OF DNA WHICH DO NOT CODE FOR PROTEIN SO IT IS NOT GENE
* GENE: SEGMENT OF DNA WHICH FORM PROTEIN
GENE
GENE X
NOT CODE FOR PROTEIN
(monocistronic) DNA / STRUCTURAL GENE / CISTRON
WHICH FORM PROTEIN
PROTEIN
PROTEIN
CISTRON / Gene
mRNA
CISTRON / Gene
tRNA
CISTRON / Gene
rRNA
SEGMENT OF DNA WHICH FORM ANY TYPE OF RNA (mRNA, tRNA, rRNA)

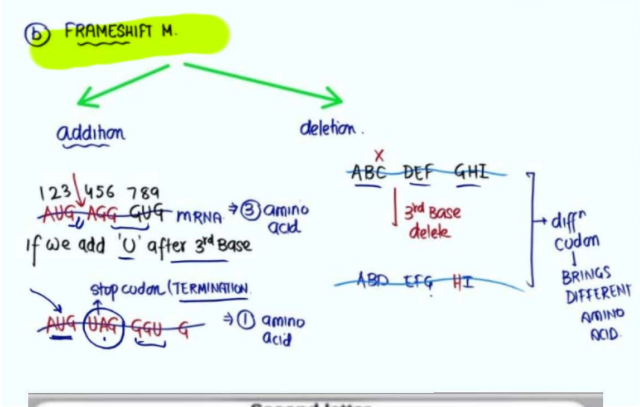

GENETIC CODE: RELATIONSHIP B/W NUCLEOTIDE SEQUENCE OF DNA & mRNA → THIS mRNA BRINGS AMINOACID (PROTEIN)
① DNA ② mRNA ③ PROTEIN
3' TAC AAA AAT ATT 5'
PR
TR
5' ATG TTT TTA TAA 3'
Q: mRNA: 4 CODON.
4th: Stop codon:
Aminoacid: 4-1 → 3
MET – Phe – Leu
NO tRNA
5' AUG UUU UUA UAA 3' mRNA
Initiation Codon/Start
TERMINATION/ STOP CODON.
Non-sense codon
CODON: 4 Codon
NUCLEOTIDE: 4x3: 12
UUU: PHENYLALANINE
UUA: LEUCINE
UAA, UAG, UGA: NO tRNA NO AMINO ACID

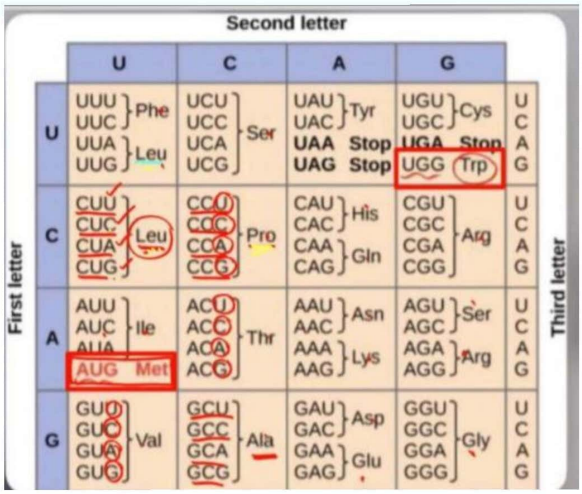

Gamow ← GENETIC CODE IS TRIPLET
* If genetic code is Single: A, U, G, C = BASE IN RNA
A mRNA
U mRNA
G mRNA
C mRNA
1 codon: 1 BASE
→ 4 CODON
(A,A)
TOTAL AMINO ACID: 20
1 CODON BRINGS 1 A.A
4 → 4 A.A
CODON ↓ Shortage
(16)
→ 4¹ → (4)
* If genetic code is double: 1 codon: 2 BASE
AU AA
AG UU
AC GG
UG CC
UA
UC
GA
GU
GC
CA
CU
CG
16 CODON → 16 A.A
CODON ↓ Shortage
(4)
→ 4² → 16
* If genetic code is TRIPLET.
4³ → 64 CODON.
3: STOP CODON.
64-3 → 61 C → 20 A.A
1 AMINO ACID CODED BY MORE THAN ONE CODON. (DEGENERACY OF GENETIC CODE)
UGG: TRYPTOPHAN
AUG: METHIONINE
EXCEPTION
NON DEGENERATE CODON.


OCHOA
POLYNUCLEOTIDE PHOSPHORYLASE (RNA Polymerase)
mRNA
ARTIFICAL SYNTHESIS OF RNA WITHOUT DNA (TEMPLATE)
KHORANA
ENZYME
mRNA homopolymer
HETROPOLYMER / COPOLYMER
MARSHAL NIRENBERG
codon decode
mRNA
amino acid
Cell free system
Phenyl alanine
Phenyl alanine


Features of genetic code
a) Triplet code
Codon: Triplet
3 Base present
b) Non ambiguous
1 codon: 1 amino acid available
Exception: GUG
Code for Two amino Acid
Ambiguous
mRNA
Valine am acid
(Initiation)
mRNA
methionine
1 CODON DO NOT CODE FOR TWO DIFF. AMINO ACID
c) Commaless
CUGUUUCUG
No punctuation.
d) Degeneracy of genetic code
1 amino acid Coded by more than one Codon.
e) Universal (NEARLY)
UU code for phenylalanine in all organism
UGA: Stop Codon (Nuclear)
UGA: Not Stop codon (mitochondria DNA)
f) Initiation Codon
methionine
AUG (Dual)
Initiation Codon
AUG (protein synthesis: start)
g) Stop codon
UAA: Ochre
UAG: Amber
UGA: Opal
h) Non overlapping
CUG UGG
2 amino acid
Exception:
AUG → Methionine
UGG → Tryptophan
Non-Degenerate Codon.


MUTATION
Change in genotype / phenotype of organism
BASE / NUCLEOTIDE SEQUENCE IN DNA
Chromosome structure
Chromosome number
CHANGE
SUDDEN CHANGE
SOURCE OF DISCONTINOUS VARIATION
CROSSING OVER: CONTINUOUS VARIATION
CHEMICALS (NITROUS ACID, MUSTARD GAS)
RADIATION (α, β, γ, X Rays)
MUTAGENS
CAUSE MUTATION
SEGMENT OF DNA → UNDERGOES MUTATION.
(MUTON)
TYPES
① GENE MUTATION
Change in nucleotides of DNA
a) SUBSTITUTION
NORMAL
CTC → CAC
TRANSITION (4 TYPES)
PURINE REPLACE BY PURINE
A → G
G → A
PYRIMIDINE REPLACE BY PYRIMIDINE
C → T
T → C
TRANSVERSION (8 TYPES)
PURINE REPLACE BY PYRIMIDINE OR PYRIMIDINE REPLACE BY PURINE
A → T
A → C
T → A
C → A
G → C
G → T
C → G
T → G


b) FRAMESHIFT M.
Addition
deletion
123 456 789
AUG AGG GUG mRNA → ③ amino acid
If we add 'U' after 3rd Base
stop codon (TERMINATION)
AUG UAG GGU G → ① amino acid
ABC DEF GHI
3rd Base delete
ABD EFG HI
diff^n codon
BRINGS DIFFERENT AMINO ACID


Second letter
U
C
A
G
First letter
Third letter
UUU Phe
UUC Phe
UUA Leu
UUG Leu
UCU Ser
UCC Ser
UCA Ser
UCG Ser
UAU Tyr
UAC Tyr
UAA Stop
UAG Stop
UGU Cys
UGC Cys
UGA Stop
UGG Trp
CUU Leu
CUC Leu
CUA Leu
CUG Leu
CCU Pro
CCC Pro
CCA Pro
CCG Pro
CAU His
CAC His
CAA Gln
CAG Gln
CGU Arg
CGC Arg
CGA Arg
CGG Arg
AUU Ile
AUC Ile
AUA Ile
AUG Met
ACU Thr
ACC Thr
ACA Thr
ACG Thr
AAU Asn
AAC Asn
AAA Lys
AAG Lys
AGU Ser
AGC Ser
AGA Arg
AGG Arg
GUU Val
GUC Val
GUA Val
GUG Val
GCU Ala
GCC Ala
GCA Ala
GCG Ala
GAU Asp
GAC Asp
GAA Glu
GAG Glu
GGU Gly
GGC Gly
GGA Gly
GGG Gly

② CHROMOSOMAL ABERRATION.

- Change in structure of Chromosome ('C')
- OBSERVED IN CANCER CELL.

★ DELETION.

A B C D E F → loss of segment/gene → A B C D

DUPLICATION

A B C D E F → A B C D E D E F (segment of chromosome: D E)

③ GENOMATIC MUTATION

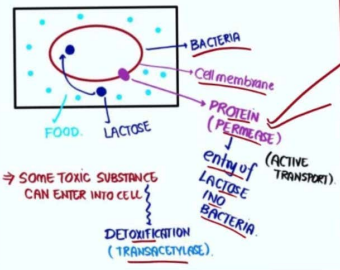
- Change in NUMBER of Chromosome

ANEUPLOIDY ✓

- loss of 'C': 45 monosomy, NULLISOMY 44
- gain of 'C': 47 TRISOMY, TETRASOMY 48

EUPLOIDY

- HAPLOIDY: 2n = 10 → n = 5.
- POLYPLOIDY: 2n: 10, 3n: 15, 4n: 20



⇒ SOME TOXIC SUBSTANCE CAN ENTER INTO CELL.

DETOXIFICATION (TRANSACETYLASE).

SECRET

⇒ LACTOSE : ENTER (पहले) → LAC OPERON ON → PERMEASE बनेगा. (बाद में)

⇒ very very low Level of enzymes present in Bacteria; Otherwise lactose will NOT ENTER INTO CELL.

REGULATION OF GENE EXPRESSION

DNA → PROTEIN

EUKARYOTES

- ★ DNA → Immature mRNA (PRIMARY TRANSCRIPT)
- ★ Immature mRNA → (PROCESSING/SPLICING) → mature mRNA
- ★ mature mRNA (NUCLEUS) → (MOVE) → mature mRNA (CYTOPLASM) ✓
- ★ mature mRNA → (TRANSLATION) → PROTEIN ✓

PROKARYOTES

⇒ DOMINANT SITE : INITIATION OF TRANSCRIPTION

WAIT

LAC OPERON (PROKARYOTES) → BACTERIA

⇒ METABOLISM OF LACTOSE (CATABOLISM) (BREAKDOWN) DISSACHARIDE.

⇒ SEGMENT OF DNA WHICH CONSIST OF GENE

- ★ REGULATOR
- ★ PROMOTER
- ★ OPERATOR
- ★ STRUCTURAL

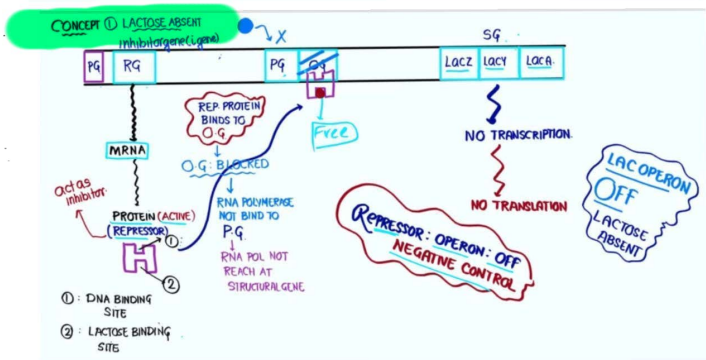
→ CONTROL

ENERGY

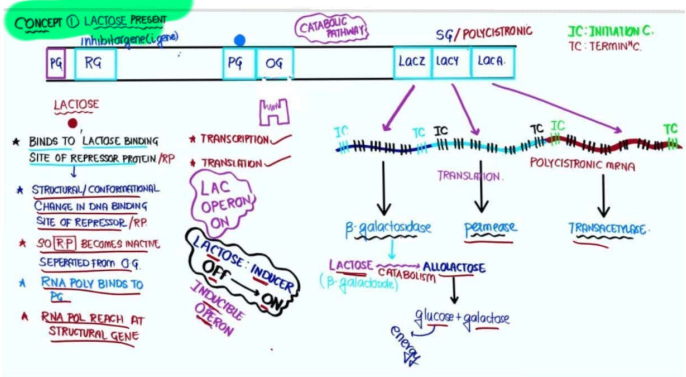
★ LACTOSE → glucose + galactose

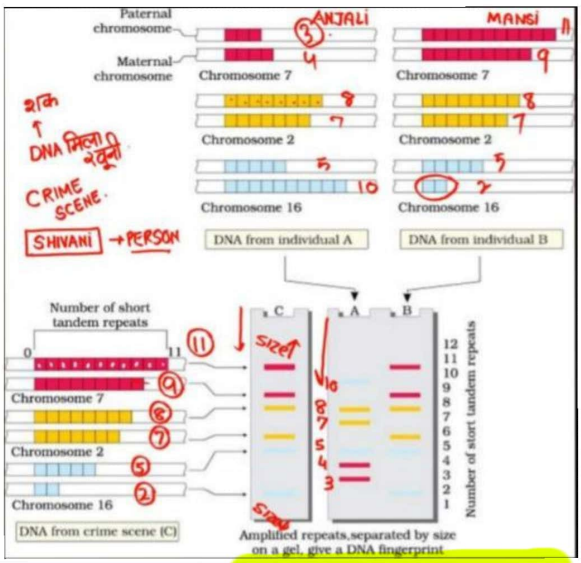
Jacob MONOD

CONCEPT ① LACTOSE ABSENT



CONCEPT ① LACTOSE PRESENT





DNA fingerprinting / DNA profiling

⇒ Human genome (Haploid set of chromosome : Length : 3.3×10^9 bp : Haploid

⇒ Diploid : length : 6.6×10^9 bp.

⇒ All Human : 99.9% DNA same

⇒ All Human : 0.1% DNA show differences/variation/polymorphism

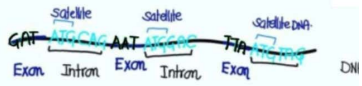
⇒ Study : 0.1% DNA

⇒ $\frac{3.3 \times 10^9 \text{ bp} \times 0.1}{100}$

⇒ 3.3×10^6 bp : variation

0.1% DNA

- Repeat after Regular interval of Time
- Satellite DNA
- Repetitive DNA
- Intronic DNA
- Not code for protein
- 3.3×10^6 bp



Q All intronic DNA are satellite DNA
a) True b) false

Q All satellite DNA are Intronic DNA
a) True b) false

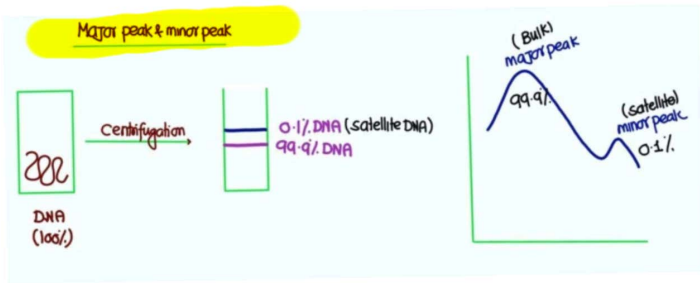
Satellite DNA

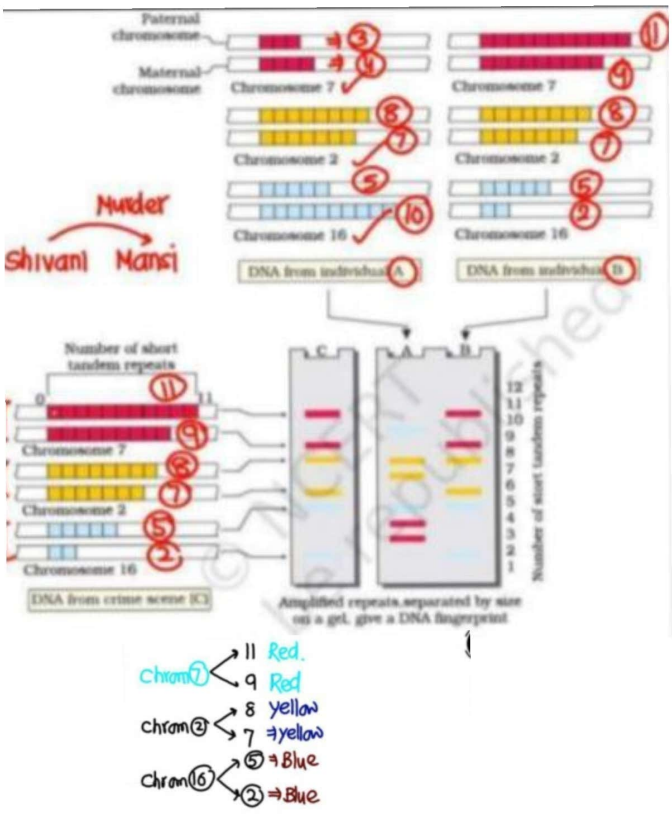
	Microsatellite (Short Tendam Repeat) 10^{-6}	Minisatellite / VNTR 10^{-3}
No. of Repeating Unit :	Four	Three
No. of Base pair in one unit	Three Base pair	10bp
Rich in	AT Rich	GC Rich
length	Short	Long
Length of Satellite	1-6 bp	10 bp - 60 bp

Variable Number of Tendam Repeat

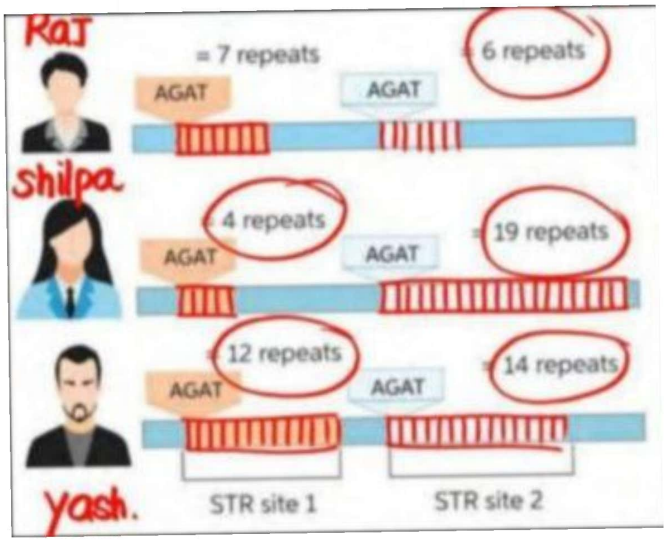
WAIT







Chrom 7 → 11 Red, 9 Red
Chrom 2 → 8 yellow, 7 → yellow
Chrom 16 → 5 → Blue, 2 → Blue



1 Repeat ⇒ AGAT (4bp) : STR/Microsatellite

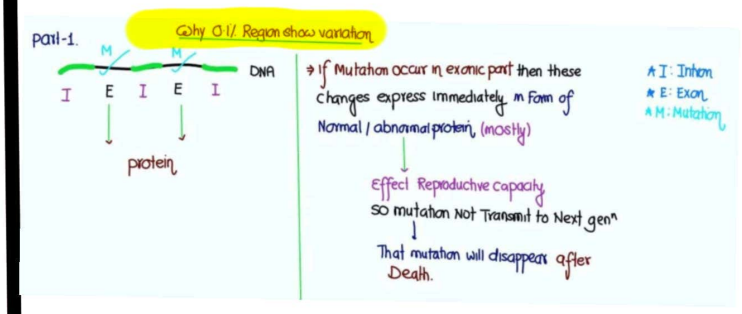
AGAT AGAT AGAT AGAT AGAT AGAT AGAT

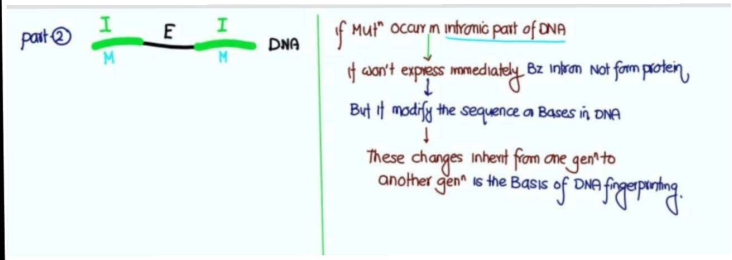
person ① ⇒ AGAT ⇒ 7+6 ⇒ (13)

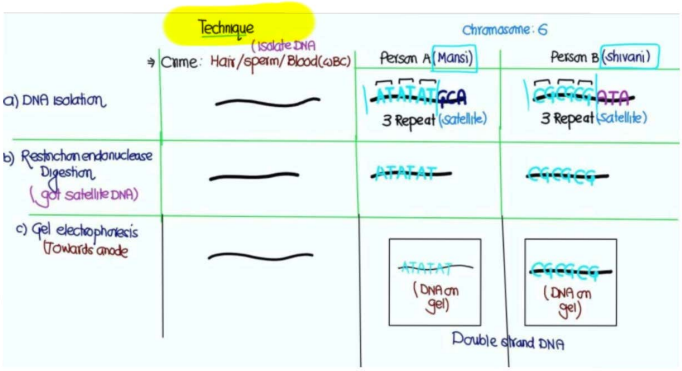
person ② ⇒ AGAT ⇒ 4+19 ⇒ (23)

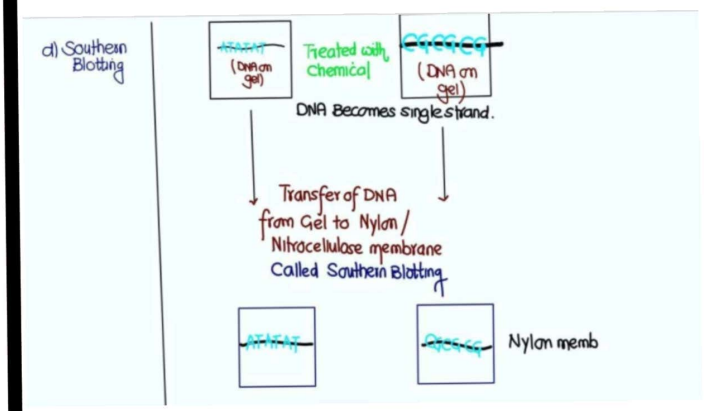
person ③ ⇒ AGAT ⇒ 12+14 = (26)

Repeating unit vary in all individual.









e) probe Hybridisation.

⇒ Artifically DNA Synthesis (single strand)

Radioactive Substance

(F) AutoRadiography (X Rays)

CRIME SCENE

DNA

Study

Find Base sequence of that DNA

ATATAT (Satellite)

Mansi

Complem. Base pairing

EST

(probe present)

Fluorescence Show

Nylon memb Shivani

No pairing probe wash out

No fluorescence.

SEMICONSERVATIVE MODE OF REPLICATION

5' G C A T 3'
3' C G T A 5'
PARENTAL DNA

semi conservative

5' G C A T 3' old/PARENT/5O/.
3' C G T A 5' NEW DNA/5O/.

5' G C A T 3' NEW DNA/5O/.
3' C G T A 5' old/PARENT/5O/.

1st Time: study: E. Coli (Bacteria)
Plants (vicia faba/Beans) → HIGHER ORGANISM.
Human cells

① MATTHEW MESSELSON & FRANKLIN STAHL (PROKARYOTE).
1958, E. Coli, $^{15}NH_4Cl$ ⇒ ^{15}N
Heavy Isotope: $^{15}NH_4Cl$ ⇒ ^{15}N
Light Isotope: $^{14}NH_4Cl$ ⇒ ^{14}N

density gradient centrifugation Technique. (Csa)

density gradient centrifugation

centrifugal FORCE DEVELOP → deposit more Cesium at Bottom

CsCl → Heavy metal

Csl density ↓ difference Csl density ↑

$N^{14}N^{14}$
$N^{15}N^{14}$ (HYBRID).
$N^{15}N^{15}$

E. coli (1 ds DNA) grow for many gen^{n} (^{14}N) Heavy Isotope.

$N^{15}N^{15}$ 1 ds DNA ($N^{15}N^{15}$) Heavy DNA

1st gen^{n} (20 minutes) (N^{14}/LIGHT ISOTOPE)

density same

1 BAND ← (ds DNA: 2)

Old ← $N^{15}N^{14}$ → New
(Old) $N^{15}N^{14}$ → New

40 MINUTES (N^{14} medium)

density same

ds DNA ② (LIGHT) $N^{14}N^{14}$ $N^{15}N^{14}$ (Hybrid) (ds DNA: ②)

(Old) $N^{15}N^{14}$ (N^{14} New) (N^{14} New) (N^{14} New)

density same

generation	Time (minutes)	Heavy DNA ($N^{15}N^{15}$)	Hybrid DNA ($N^{15}N^{14}$)	Light DNA ($N^{14}N^{14}$)
1st	20	0	2	0
2nd	40	0	2	0+2 = (2)
3rd	60	0	2	2+4 = (6)
4th	80	0	2	6+8 = (14)

SHORT CUT

$N^{15}N^{15}$ → N^{14} (1st gen^{n})

$N^{15}N^{14}$ $N^{15}N^{14}$ → N^{14} (2nd gen^{n}).

$N^{15}N^{14}$ $N^{14}N^{14}$ $N^{15}N^{14}$ $N^{14}N^{14}$

② Taylor.
Beans/vicia faba, (EUKARYOTE).
Radioactive Thymidine

① Both strand of Bacteria have Thymidine grow for Two gen^{n} in NON-Radioactive Thymidine

TT → NT
T NT T NT → NT
T NT NT NT T NT NT NT

③ CAIRNS
Radioactive Thymidine
E. Coli (PROKARYOTE)

PACKAGING OF DNA IN EUKARYOTES

NUCLEUS: 10^{-6}m.
DNA: 2.2m.

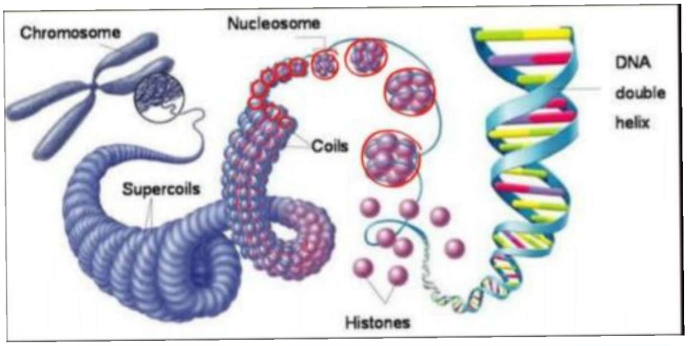
(PHOSPHATE) DNA: Negative charge
HP: HISTONE PROTEIN

HP = 4 TYPES
1 TYPE: ②
⇒ OCTAMER OF HP.
⇒ Positively charge

200bp
1.75 TURN OF DNA AROUND HP
IN
H_1

HISTONE P
⇒ BASIC P
⇒ POSITIVELY CHARGE AMINO ACID (ARGININE, LYSINE)
⇒ Packaging of DNA (Low level): Starting.
⇒ IN: NUCLEOSOME
DNA octamer of HP
⇒ H_1: NOT A PART OF 'N'
: CONNECT TWO 'N'

PACKAGING OF DNA IN EUKARYOTES

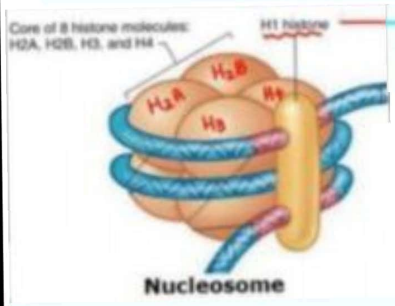


A DNA + HP (Basic) → NUCLEOSOME → MORE PACKAGING MORE NUCLEOSOME + NON HIST. PROTEIN (SCAFFOLD PROTEIN) (ACIDIC)/ HIGH LEVEL → Chromatin (INTERPHASE)

LOW LEVEL

Chromatin Condensed chromosome

COND. CHROMOSOME (METAPHASE)



① How many Beads (Nucleosome) in eukaryotic DNA.

Nucleosome (Bead)
Bead on string

No. of Nucleosome : $\frac{6.6\times10^{9}\text{ bp}}{200\text{ bp}}$

⇒ 3.3×10^{7}

② If length of E. coli: 1.36mm
Can you calculate no. of Base pair in E. coli

⇒ length = no. of Bp x distance B/W bp

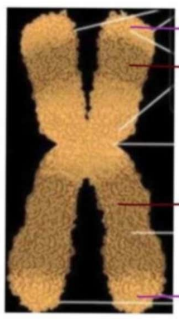
⇒ $\frac{1.36\times10^{-3}\text{m}}{0.34\times10^{-9}\text{m}}$: no of Base

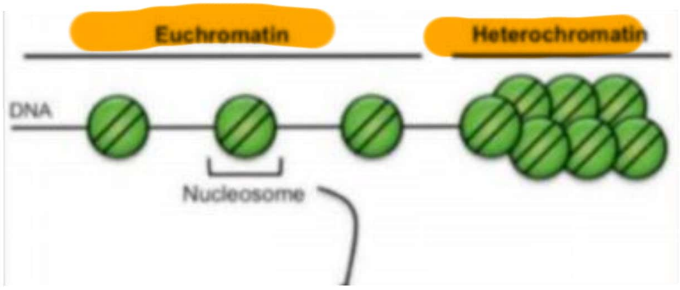
⇒ 4×10^{6} bp

Chromatin

Euchromatin
⇒ loosely packed.
⇒ LIGHTLY STAIN
DNA (TRANSCRIPTIONALLY ACTIVE)
mRNA

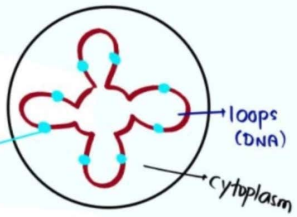
Heterochromatin
⇒ TIGHTLY PACKED.
⇒ DARKLY STAIN.
DNA (TRANSCRIPTIONALLY INACTIVE)
NO mRNA





Packaging in prokaryotes

- Well defined NUCLEUS ABSENT
- ⇒ HISTONE PROTEIN: ABSENT
  IT DOESNT MEAN NUCLEUS SCATTERED IN CYTOPLASM
- ⇒ PACKAGING: NON HISTONE PROTEIN / POLYAMINE (BASIC)



TRANSLATION mRNA → PROTEIN — INITIATION, ELONGATION, TERMINATION

AMINOACYLATION OF tRNA / Charging of tRNA

a) ACTIVATION OF AMINO ACID

AMINOACID + ATP → AMINOACYL AMP + PP
(INACTIVE) — ACTIVE FORM OF amino acid

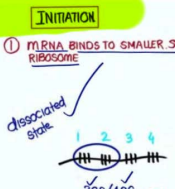
b) CHARGING OF tRNA

AMINOACYL AMP + tRNA → CHARGED tRNA: AMINOACYL tRNA + AMP

This Reactions Catalysed By Amino acyl tRNA Synthetase.

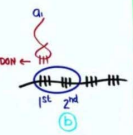
INITIATION

① mRNA BINDS TO SMALLER SUBUNIT OF RIBOSOME

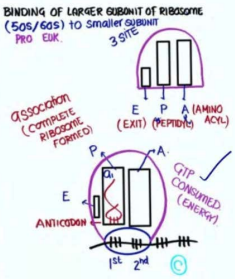


I : INITIATION C (AUG)
4 : TERMINATION C

② 1st t-RNA (initiator) carry 1st amino acid.
P : FORMYL METHIONINE
E : METHIONINE



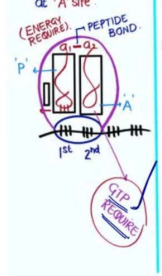
③ BINDING OF LARGER SUBUNIT OF RIBOSOME (50S/60S) TO SMALLER SUBUNIT
PRO EUK



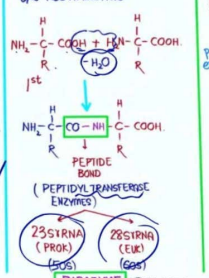
NOTE: SOME PROTEINS (INITIATION FACTORS) INVOLVED
PROK: 3 → IF-1, IF-2, IF-3
EUK: 9

ELONGATION

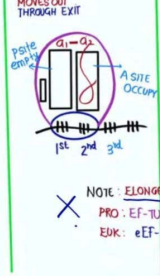
⇒ BINDING OF 2ND tRNA AT 'A' SITE



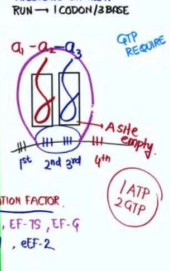
⇒ PEPTIDE BOND FORMATION B/w TWO AMINO ACID



⇒ tRNA at P SITE DEGRADED & MOVES OUT THROUGH EXIT



Translocation.
⇒ movement / sliding of RIBOSOME on MRNA
RUN → 1 CODON / 3 BASE

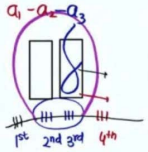


1 ATP 2 GTP

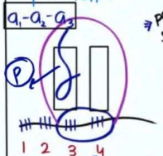
NOTE: ELONGATION FACTOR
PRO: EF-TU, EF-TS, EF-G
EUK: eEF-1, eEF-2

TERMINATION

tRNA OF P SITE DEGRADE & MOVES OUT THROUGH 'E' SITE



RIBOSOME AGAIN MOVE
growing polypeptide



UAA (stop codon: no tRNA)

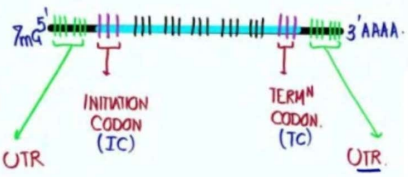
⇒ PROTEIN SYNTHESIS STOP
LAST AMINO ACID/ growing polypeptide chain at P site

RELEASING FACTOR (PROTEIN) → SEPERATE POLYPEPTIDE CHAIN FROM tRNA.
RF-1
RF-2

GTP CONSUMED

★ 1 ATP: Activation of amino acid
★ 1 GTP: SUDING.
★ 1 GTP: tRNA BIND AT A SITE

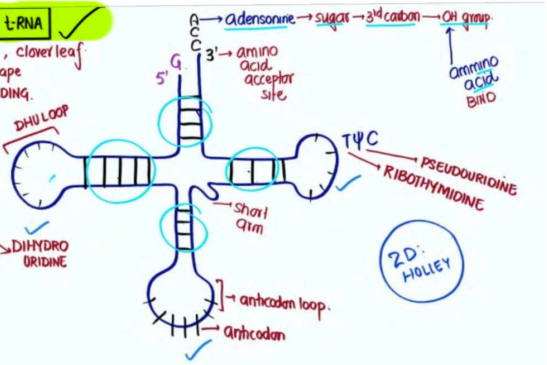
mRNA

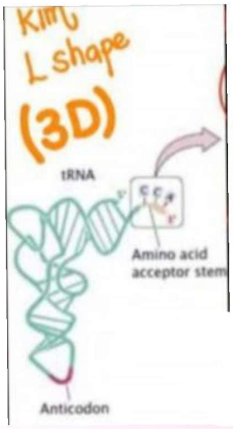


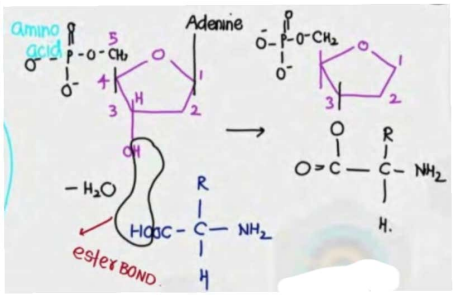
UTR: UNTRANSLATED REGION IN mRNA BEFORE IC & after TC
: PROTEIN NOT FORMED BUT THEY MAKES TRANSLATION EFFICIENT

t-RNA

- 2D/SECONDARY ST: HOLLEY, clover leaf
- 3D: KIM., Inverted L Shape
- single standard: BUT FOLDING. SO it appear as double strand.
- 5' end: GUANINE.
- 3' end: CCA Sequence.
- ⇒ Some abnormal Base present Pairing NOT POSSIBLE: LOOP ARE FORMED







	mRNA	tRNA	rRNA
Synthesis	RNA POL-II	RNA POL-III	RNA POL-I
Location	NUCLEUS	NUCLEUS	NUCLEOLUS
Adapter RNA	×	adapter for amino acid & codon. a1 Codon mRNA	×
Soluble RNA	×	SOLUBLE IN NACL	×
Size	small	Smallest	Largest
Percentage	1-5%	10-15%	50-80%